

Computer Programming 143 – Lecture 21

Pointers V

Faculty of Engineering



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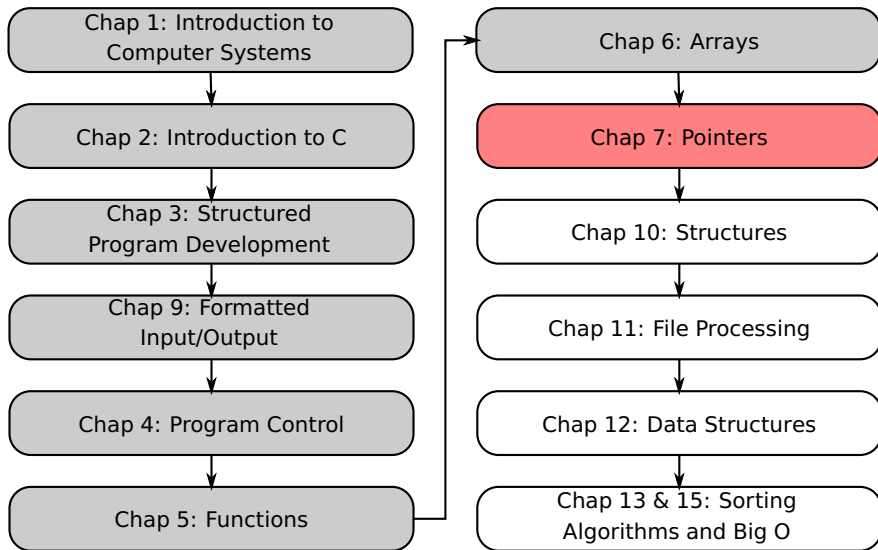
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Module Overview



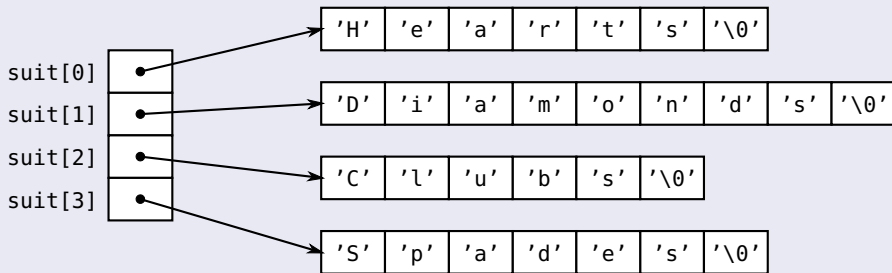
1 Arrays of Pointers (7.10)

2 Design Example: Card Shuffling and Dealing (7.11)

Array of strings

```
const char *suit[ 4 ] = { "Hearts", "Diamonds", "Clubs", "Spades" };
```

- Declares an array of 4 pointers to type char
- Initialises 4 strings and assigns the addresses of the first characters to the 4 pointers



7.11 Card Shuffling and Dealing I

Problem statement

Design and implement an algorithm that shuffles and deals a 52-card deck

7.11 Card Shuffling and Dealing II

Deck representation

```
int deck[ 4 ][ 13 ];
```

		Ace	Two	Three	Four	Five	Six	Seven	Eight	Nine	Ten	Jack	Queen	King
		0	1	2	3	4	5	6	7	8	9	10	11	12
Hearts	0													
Diamonds	1													
Clubs	2													
Spades	3													

deck[2][12] represents the King of Clubs

The value stored in deck[2][12], represents the card's unique position in the deck

7.11 Card Shuffling and Dealing III

Empty Deck representation

		Ace	Two	Three	Four	Five	Six	Seven	Eight	Nine	Ten	Jack	Queen	King
		0	1	2	3	4	5	6	7	8	9	10	11	12
Hearts	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Diamonds	1	0	0	0	0	0	0	0	0	0	0	0	0	0
Clubs	2	0	0	0	0	0	0	0	0	0	0	0	0	0
Spades	3	0	0	0	0	0	0	0	0	0	0	0	0	0

7.11 Card Shuffling and Dealing IV

Ordered Deck representation

		Ace	Two	Three	Four	Five	Six	Seven	Eight	Nine	Ten	Jack	Queen	King
		0	1	2	3	4	5	6	7	8	9	10	11	12
Hearts	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Diamonds	1	14	15	16	17	18	19	20	21	22	23	24	25	26
Clubs	2	27	28	29	30	31	32	33	34	35	36	37	38	39
Spades	3	40	41	42	43	44	45	46	47	48	49	50	51	52

7.11 Card Shuffling and Dealing V

Shuffled Deck representation

		Ace	Two	Three	Four	Five	Six	Seven	Eight	Nine	Ten	Jack	Queen	King
		0	1	2	3	4	5	6	7	8	9	10	11	12
Hearts	0	26	51	16	44	36	15	50	46	3	38	52	40	10
Diamonds	1	11	19	34	27	1	20	47	21	25	4	32	18	31
Clubs	2	6	35	8	9	37	2	49	29	22	13	5	42	33
Spades	3	12	7	23	45	24	41	28	14	39	30	48	43	17

Top-level pseudocode

Shuffle and deal 52 cards

First refinement

Initialise the constant array of suit names
Initialise the constant array of face names
Initialise the deck array of card positions
Shuffle the deck
Deal 52 cards

Second refinement

Initialise the constant array of suit names

Initialise the constant array of face names

Initialise the deck array of card positions

For each of the 52 ordinal positions in the empty deck

Place ordinal position number in randomly selected unoccupied slot of deck

For each of the 52 ordinal positions in the shuffled deck

Find ordinal position number in deck array and print face and suit of card

Third refinement

Initialise the suit array

Initialise the face array

Initialise the deck array

For each of the 52 ordinal positions in the empty deck

Do

Choose slot of deck randomly

While slot of deck is not empty

Place card number in chosen slot of deck

For each of the 52 ordinal positions in the shuffled deck

For each slot of deck array

If slot contains desired card number

Print the face and suit of the card

```

#include <stdio.h>
#include <stdlib.h>
#include <time.h>

// prototypes
void shuffle( int wDeck[][ 13 ] );
void deal( const int wDeck[][ 13 ], const char *wFace[],
          const char *wSuit[] );

int main( void )
{
    // initialise suit array
    const char *suit[ 4 ] = { "Hearts", "Diamonds", "Clubs", "Spades" };
    // initialise face array
    const char *face[ 13 ] = { "Ace", "Two", "Three", "Four", "Five",
        "Six", "Seven", "Eight", "Nine", "Ten", "Jack", "Queen", "King" };
    // initialise deck array
    int deck[ 4 ][ 13 ] = {{ 0 }};
    srand( time( NULL ) ); // seed random-number generator
    shuffle( deck ); // shuffle the deck
    deal( deck, face, suit ); // deal the deck
    return 0; // indicates successful termination
} // end main

```

```

// shuffle cards in deck
void shuffle( int wDeck[][ 13 ] ) {
    int row; // row number
    int column; // column number
    int card; // counter

    // for each of the 52 ordinal positions, choose slot of deck randomly
    for ( card = 1; card <= 52; card++ ) {
        // choose new random location until unoccupied slot found
        do {
            row = rand() % 4;
            column = rand() % 13;
        } while ( wDeck[ row ][ column ] != 0 ); // end do...while
        // place card number in chosen slot of deck
        wDeck[ row ][ column ] = card;
    } // end for
} // end function shuffle

```

```

// deal cards in deck
void deal( const int wDeck[][ 13 ], const char *wFace[],
          const char *wSuit[] ) {
    int card, row, column; // card, row & column counters
    // deal each of the 52 ordinal positions
    for ( card = 1; card <= 52; card++ ) {
        for ( row = 0; row <= 3; row++ ) { // loop rows
            for ( column = 0; column <= 12; column++ ) { // loop columns
                // if slot contains current card, display card
                if ( wDeck[ row ][ column ] == card ) {
                    if (card % 2 == 0){ // 2-column format
                        printf("%5s of %-8s%c",wFace[column],wSuit[row],'\n');
                    }
                    else {
                        printf("%5s of %-8s%c",wFace[column],wSuit[row],'\t');
                    } // end else
                } // end if
            } // end for
        } // end for
    } // end for
} // end function deal

```

Example output

Seven of Diamonds	Jack of Hearts
Four of Spades	Nine of Hearts
Ace of Clubs	Three of Diamonds
Two of Spades	Ten of Clubs
King of Diamonds	Six of Hearts
Queen of Spades	Five of Clubs
Eight of Diamonds	Ace of Hearts
Nine of Diamonds	Three of Spades
Five of Diamonds	Ten of Hearts
Two of Clubs	Eight of Hearts
Jack of Spades	King of Hearts
Six of Clubs	Queen of Clubs
Four of Diamonds	Two of Diamonds
Seven of Spades	Eight of Clubs
Nine of Spades	King of Spades
Ten of Diamonds	Six of Diamonds
Five of Hearts	Ace of Diamonds
Three of Hearts	Jack of Diamonds
Queen of Hearts	Seven of Hearts
Four of Clubs	King of Clubs
Two of Hearts	Jack of Clubs
Ten of Spades	Nine of Clubs
Eight of Spades	Ace of Spades
Seven of Clubs	Six of Spades
Queen of Diamonds	Four of Hearts
Three of Clubs	Five of Spades

Today

Pointers V

- Arrays of pointers
- Design example: card shuffling and dealing

Next lecture

- Structures

- 1 Study Sections 7.10-7.11 in Deitel & Deitel
- 2 Do Exercise 7.21